Adding and Subtracting Big Numbers

Line up the places. Start with the ones.

Add: carry a new ten, hundred or thousand

	1	1	1	
	3	4	8	6
+	2	7	9	5
	6	2	8	1

Ones: 6 + 5 = 11. Write 1, carry 1 ten.

Tens: 1 + 8 + 9 = 18. Write 8, carry 1.

Hundreds: 1 + 4 + 7 = 12. Write 2, carry 1.

Thousands: 1 + 3 + 2 = 6.

Subtract: break a bigger place when you need to

	5	12	4	13
	~~6~~	~~2~~	~~5~~	~~3~~
−	2	7	1	8
	3	5	3	5

Ones: 3 is less than 8. Break a ten: 13 − 8 = 5.

Tens: 4 − 1 = 3.

Hundreds: 2 is less than 7. Break a thousand: 12 − 7 = 5. Thousands: 5 − 2 = 3.

Your turn!

	☐	☐	1	
	4	5	1	8
+	1	3	6	7
	5	8	8	5

For teachers

Goal: add and subtract numbers to 9,999 in columns (4.NBT.4), the same way as packets 13, 14 and 29 with one more place. Say “Line up the places. Start with the ones. Do I need to carry or break?”

Adding and Subtracting Big Numbers: Practice

Line up the places. Start with the ones.

Part 1 Add with carry boxes Write each carried 1 in its box.

1)

	1	☐	1	
	2	7	4	6
+	1	5	3	8
	4	2	8	4

2)

	☐	1	1	
	4	1	8	5
+	3	4	2	7
	7	6	1	2

3)

	1	1	1	
	3	6	5	9
+	2	8	7	3
	6	5	3	2

4)

	1	1	1	
	5	3	0	8
+	2	6	9	4
	8	0	0	2

5)

	1	1	1	
	1	8	7	4
+	6	2	5	8
	8	1	3	2

6)

	1	1	1	
	6	4	2	7
+	1	5	8	6
	8	0	1	3

Part 2 Subtract with boxes Cross out and write the new digit in the box.

7)

	6	12	4	13
	~~7~~	~~2~~	~~5~~	~~3~~
−	2	6	1	8
	4	6	3	5

8)

	4	13	11	10
	~~5~~	~~4~~	~~2~~	~~0~~
−	1	8	7	3
	3	5	4	7

Adding and Subtracting Big Numbers: Practice (continued)

Part 2 Subtract with boxes (continued)

9)

	7	12	11	
	~~8~~	~~3~~	~~1~~	6
−	4	5	9	2
	3	7	2	4

10)

	5	10		
	~~6~~	~~0~~	4	5
−	2	7	3	1
	3	3	1	4

11)

	8	11	7	12
	~~9~~	~~1~~	~~8~~	~~2~~
−	3	4	7	5
	5	7	0	7

12)

	3	16	12	11
	~~4~~	~~7~~	~~3~~	~~1~~
−	2	8	5	6
	1	8	7	5

Part 3 Add No boxes this time. Carry above the digits.

13)

	1	1	1	
	3	7	5	8
+	2	4	6	5
	6	2	2	3

14)

	1	1	1	
	1	9	4	6
+	5	3	8	7
	7	3	3	3

15)

	1		1	
	4	6	2	9
+	3	8	5	4
	8	4	8	3

16)

	1	1	1	
	2	5	8	7
+	6	5	1	9
	9	1	0	6

Adding and Subtracting Big Numbers: Practice (continued)

Part 4 Subtract No boxes this time.

17)

	7	10	11	14
	~~8~~	~~1~~	~~2~~	~~4~~
−	3	5	6	7
	4	5	5	7

18)

		2	14	12
	6	~~3~~	~~5~~	~~2~~
−	4	1	7	8
	2	1	7	4

19)

	6	14	10	16
	~~7~~	~~5~~	~~1~~	~~6~~
−	2	8	3	9
	4	6	7	7

20)

	8	11	15	13
	~~9~~	~~2~~	~~6~~	~~3~~
−	5	4	7	8
	3	7	8	5

Part 5 Across zeros Break the thousand first when the next places are 0.

21)

	4	9	9	13
	~~5~~	~~0~~	~~0~~	~~3~~
−	2	4	7	8
	2	5	2	5

22)

	6	9	9	10
	~~7~~	~~0~~	~~0~~	~~0~~
−	3	6	2	4
	3	3	7	6

23)

	3	9	9	16
	~~4~~	~~0~~	~~0~~	~~6~~
−	1	8	3	7
	2	1	6	9

24)

	7	9	9	10
	~~8~~	~~0~~	~~0~~	~~0~~
−	5	2	5	9
	2	7	4	1

Adding and Subtracting Big Numbers: Practice (continued)

Part 6 Add No help this time. Write them in columns on scrap paper if it helps.

25) 4,576 + 2,849 = 7,425

26) 3,218 + 5,694 = 8,912

27) 6,735 + 1,987 = 8,722

28) 2,468 + 3,759 = 6,227

29) 5,189 + 2,936 = 8,125

30) 1,897 + 4,365 = 6,262

31) 7,342 + 1,789 = 9,131

32) 3,906 + 4,597 = 8,503

Part 7 Subtract No help this time.

33) 7,425 − 2,849 = 4,576

34) 6,031 − 3,785 = 2,246

35) 9,104 − 4,567 = 4,537

36) 5,280 − 1,694 = 3,586

37) 8,352 − 6,478 = 1,874

38) 4,019 − 2,563 = 1,456

39) 7,600 − 3,825 = 3,775

40) 6,243 − 1,876 = 4,367

Part 8 Remember? Problems from earlier packets.

41) 5,628 to the nearest thousand = 6,000

42) in 4,217 the 2 is worth 200

43) 450 g + 300 g = 750 g

44) 1,000 less than 6,250 = 5,250

45) each picture = 10: 7 pictures = 70

46) 2¾ inches = 11 quarter inches

Adding and Subtracting Big Numbers: Practice (continued)

Part 9 Word problems Draw a bar if it helps.

47) A school collected 2,478 cans in fall and 3,865 cans in spring. How many cans in all?

Number sentence 2,478 + 3,865 = 6,343

Answer 6,343 cans

Draw or show your work

48) A mountain trail is 5,280 feet long. Kim has walked 2,645 feet. How far is left?

Number sentence 5,280 − 2,645 = 2,635

Answer 2,635 feet

Draw or show your work

49) A zoo had 4,906 visitors on Saturday and 3,958 on Sunday. How many more came on Saturday?

Number sentence 3,958 + 948 = 4,906

Answer 948 visitors

Draw or show your work

50) A truck carries 1,875 pounds. It picks up 2,340 more. How heavy is the load now?

Number sentence 1,875 + 2,340 = 4,215

Answer 4,215 pounds

Draw or show your work

51) Challenge: A store had 6,000 pencils. It sold 1,250 on Monday and 2,375 on Tuesday. How many are left?

Number sentence 1,250 + 2,375 = 3,625, 6,000 − 3,625 = 2,375

Answer 2,375 pencils

Draw or show your work

Check-In: Adding and Subtracting Big Numbers

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Add in columns

1)

$$\begin{array}{r} {\scriptstyle 1} \\ 3396 \\ +\ 3363 \\ \hline 6759 \end{array}$$

2)

$$\begin{array}{r} {\scriptstyle 1} \\ 2713 \\ +\ 2942 \\ \hline 5655 \end{array}$$

Part 2 Subtract in columns

3)

$$\begin{array}{r} 9275 \\ -\ 3204 \\ \hline 6071 \end{array}$$

4)

$$\begin{array}{r} 7496 \\ -\ 1376 \\ \hline 6120 \end{array}$$

Part 3 Add

5) $4{,}110 + 2{,}993 =$ **7,103**

6) $6{,}505 + 3{,}727 =$ **10,232**

Part 4 Subtract

7) $5{,}664 - 4{,}459 =$ **1,205**

8) $9{,}723 - 4{,}459 =$ **5,264**